Lecture 10. Simulation

Inverse transform, rejection method, discrete distributions, variance reduction

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10.1 Random numbers and simulation

When a probability or expectation is too complicated to compute, we can estimate it by **simulation**: generate many independent realizations of the experiment and use the strong law of large numbers. Everything is built from **random numbers**, independent $U(0,1)$ variables (in practice produced by deterministic pseudorandom generators). For instance, $\lfloor nU \rfloor + 1$ is uniform on $\{1, \dots, n\}$.

Example 10.1 (A random permutation)

To generate a uniformly random permutation of $1, \dots, n$: start from any ordering $P_1, \dots, P_n$; for $k = n, n-1, \dots, 2$, choose I uniform on $\{1, \dots, k\}$ and swap P_I and P_k. Position n receives each element with probability $1/n$, then position $n-1$ each remaining element with probability $1/(n-1)$, and so on, so every permutation has probability $1/n!$. The first r positions filled give a random r-subset, which is how subjects are randomly assigned to treatments.

10.2 Simulating continuous random variables

Motivation. Computers produce uniform random numbers; every other distribution must be built from them. The inverse transform and rejection methods are the two general techniques.

10.2.1 The inverse transformation method

Proposition 10.1

If F is a continuous cdf and $U \sim U(0,1)$, then $X = F^{-1}(U)$ has cdf F.

Proof. $P(F^{-1}(U) \le x) = P(U \le F(x)) = F(x)$, since F is increasing. ∎

Example 10.2

(a) Exponential: $F(x) = 1 - e^{-\lambda x}$ gives $X = -\frac{1}{\lambda}\log(1-U)$, or equivalently $-\frac{1}{\lambda}\log U$, since $1-U$ is also uniform.
(b) Gamma(n, λ) for integer n: add n independent exponentials, $X = -\frac{1}{\lambda}\log(U_1 U_2 \cdots U_n)$, which needs only one logarithm.

10.2.2 The rejection method

Suppose we can simulate from a density g, and $f(y)/g(y) \le c$ for all y.

Theorem 10.2 (Rejection method)

Repeat: generate $Y \sim g$ and an independent $U \sim U(0,1)$, until $U \le f(Y)/\{cg(Y)\}$; then set $X = Y$. The resulting X has density f, and the number of iterations is geometric with mean c.

Proof. In one iteration, $P(\text{accept}) = \int P(U \le \frac{f(y)}{cg(y)}) g(y)\, dy = \int \frac{f(y)}{c}\, dy = \frac{1}{c}$, and

$$P(Y \le x \mid \text{accept}) = c \int_{-\infty}^{x} \frac{f(y)}{cg(y)} g(y)\, dy = F(x).$$

Iterations are independent, so the number needed is geometric with success probability $1/c$, and the accepted value has the conditional law just computed. ■

Example 10.3 (Normal from exponentials)

$|Z|$ has density $f(x) = \frac{2}{\sqrt{2\pi}} e^{-x^2/2}$, $x > 0$. With $g(x) = e^{-x}$,

$$\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}}\, e^{x - x^2/2} \le \sqrt{\frac{2e}{\pi}} = c \approx 1.32,$$

the maximum being at $x = 1$, and $\frac{f(x)}{cg(x)} = \exp\{-(x-1)^2/2\}$. So: generate $Y = -\log U_1$, accept if $U_2 \le e^{-(Y-1)^2/2}$ (equivalently if an independent exponential exceeds $(Y-1)^2/2$), and finally attach a random sign. About 1.32 iterations are needed per normal.

Example 10.4 (The polar method)

By Example 6.14, if $R^2 \sim \text{Exp}(1/2)$ and $\Theta \sim U(0, 2\pi)$ are independent, then $R\cos\Theta$, $R\sin\Theta$ are independent standard normals. To avoid computing sines and cosines, generate $V_i = 2U_i - 1$ $(i = 1, 2)$ until $S = V_1^2 + V_2^2 \le 1$; then (V_1, V_2) is uniform on the unit disc, S is uniform and independent of the angle, and $\cos\Theta = V_1/\sqrt{S}$, $\sin\Theta = V_2/\sqrt{S}$. With $R^2 = -2\log S$,

$$X = V_1\sqrt{\frac{-2\log S}{S}}, \qquad Y = V_2\sqrt{\frac{-2\log S}{S}}$$

are independent standard normals. The acceptance probability is $\pi/4$.

A χ^2_n variable can be simulated as $\sum Z_i^2$, or for even n as $-2\log(U_1 \cdots U_{n/2})$ (a Gamma$(n/2, 1/2)$), adding one Z^2 for odd n.

10.3 Simulating from discrete distributions

For $P(X = x_j) = p_j$, set $X = x_j$ if $\sum_{i<j} p_i \le U < \sum_{i \le j} p_i$; this happens with probability p_j. It is the discrete inverse transform.

Example 10.5

(a) Geometric: $X = j$ iff $1 - q^{j-1} \le U < 1 - q^j$ $(q = 1 - p)$, i.e. $X = 1 + \lfloor \log U / \log q \rfloor$ (using $1 - U$ in place of U).
(b) Binomial: $X = \sum_{i=1}^n \mathbb{1}\{U_i < p\}$.
(c) Poisson(λ): with $N(1)$ the number of events of a rate-λ process in $[0, 1]$, and interarrival times $-\log U_i/\lambda$,

$$N = \max\Big\{n : \prod_{i=1}^{n} U_i \ge e^{-\lambda}\Big\}$$

is Poisson(λ): multiply random numbers until the product drops below $e^{-\lambda}$, and subtract 1 from the count.

10.4 Variance reduction

To estimate $\theta = E[X]$ we average n independent copies, with error of order $\sqrt{\mathrm{Var}(X)/n}$. Reducing the variance of what is averaged is as good as increasing n.

10.4.1 Antithetic variables

If X_1, X_2 have the same distribution, $\mathrm{Var}(\frac{X_1+X_2}{2}) = \frac{1}{4}\{2\,\mathrm{Var}(X_1) + 2\,\mathrm{Cov}(X_1, X_2)\}$, which is smaller than for independent copies when $\mathrm{Cov}(X_1, X_2) < 0$. If $X_1 = h(U_1, \dots, U_m)$ with h monotone in each argument, then $X_2 = h(1 - U_1, \dots, 1 - U_m)$ has the same distribution and (as can be shown) $\mathrm{Cov}(X_1, X_2) \le 0$. So each set of random numbers is used twice, once "reversed".

10.4.2 Variance reduction by conditioning

By the conditional variance formula (Proposition 7.6), $\mathrm{Var}(E[X \mid Y]) \le \mathrm{Var}(X)$, and $E[X \mid Y]$ has the same mean θ. So, if $E[X \mid Y]$ can be computed, simulate Y and average $E[X \mid Y]$ instead of X.

Example 10.6 (Estimating π)

With V_1, V_2 independent $U(-1, 1)$ and $I = \mathbb{1}\{V_1^2 + V_2^2 \le 1\}$, $E[I] = \pi/4$ and $\mathrm{Var}(I) = \frac{\pi}{4}(1 - \frac{\pi}{4}) \approx 0.1685$. Conditioning on V_1,

$$E[I \mid V_1] = P(V_2^2 \le 1 - V_1^2 \mid V_1) = \sqrt{1 - V_1^2},$$

so we may average $\sqrt{1 - U^2}$ over random numbers U (the sign of V_1 is irrelevant): $\mathrm{Var}(\sqrt{1 - U^2}) = \frac{2}{3} - \frac{\pi^2}{16} \approx 0.0498$, a 3.4-fold reduction. Since $\sqrt{1 - u^2}$ is monotone, antithetic pairs $\frac{1}{2}\{\sqrt{1 - U^2} + \sqrt{1 - (1 - U)^2}\}$ do better still: their variance is about 0.0069 per pair, against 0.0249 for two independent values.

10.4.3 Control variates

Suppose Y is simulated alongside X and $\mu_Y = E[Y]$ is known. Then $X + c(Y - \mu_Y)$ also has mean θ, and its variance $\mathrm{Var}(X) + c^2\,\mathrm{Var}(Y) + 2c\,\mathrm{Cov}(X, Y)$ is minimized by

$$c^* = -\frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(Y)}, \qquad \mathrm{Var}(X + c^*(Y - \mu_Y)) = \mathrm{Var}(X)(1 - \rho(X, Y)^2).$$

The more strongly Y is correlated with X, the larger the gain; in practice c^* is estimated from the simulated pairs. In Example 10.6, $Y = U^2$ with $E[Y] = 1/3$ is a natural control variate for $\sqrt{1 - U^2}$.

10.5 Selected exercises

Motivation. Exercises 10.1–10.13 are Ross's Chapter 10 problems (book numbering); they prove that the simulation recipes of this lecture are correct or optimal. Exercise 10.A is added to justify the claim, made above without proof, that antithetic pairs are negatively correlated.

Exercise 10.1 (Random permutations by insertion)

Set $P(1) = 1$. For $k = 2, \dots, n$, generate a random number U, let $J = \lfloor kU \rfloor + 1$, and set $P(k) = P(J)$ and then $P(J) = k$ (so for $J = k$ simply $P(k) = k$). Show that after step k, $(P(1), \dots, P(k))$ is a uniformly random permutation of $1, \dots, k$.

Solution.

Step 1. In step k the algorithm puts k into position J and moves the element previously at position J to the new position k; J is uniform on $\{1, \dots, k\}$ and independent of the earlier steps.

Step 2. Let Φ map a pair (σ, j), with σ a permutation of $1, \dots, k-1$ and $j \in \{1, \dots, k\}$, to the result τ of this operation. Φ is a bijection onto the permutations of $1, \dots, k$: given τ, j is the position of k in τ, and σ is recovered by putting $\tau(k)$ back into position j (if $j < k$) and deleting position k.

Step 3. (Induction.) After step 1 the claim is trivial. If after step $k-1$ each σ has probability $1/(k-1)!$, then by independence each pair (σ, j) has probability $\frac{1}{(k-1)!} \cdot \frac{1}{k}$, and by Step 2 each permutation τ of $1, \dots, k$ arises from exactly one pair. So each τ has probability $1/k!$. ■

Exercise 10.7 (The maximum of random numbers)

Let $F(x) = x^n$ for $0 < x < 1$.
(a) Simulate from F using a single random number.
(b) Show that $\max(U_1, \dots, U_n)$ has distribution F for independent random numbers $U_1, \dots, U_n$, and so give a second method.

Solution.

(a) F is continuous and strictly increasing on $(0, 1)$ with $F^{-1}(u) = u^{1/n}$, so $X = U^{1/n}$ has distribution F by Proposition 10.1: $P(U^{1/n} \le x) = P(U \le x^n) = x^n$.

(b) $\max_i U_i \le x$ iff every $U_i \le x$, so by independence $P(\max_i U_i \le x) = \prod_i P(U_i \le x) = x^n$. Thus $\max(U_1, \dots, U_n)$ is a second (more expensive) way to generate from F. ■

Exercise 10.10 (Choosing the exponential envelope)

In Example 10.3, use the envelope $g_\lambda(x) = \lambda e^{-\lambda x}$, $x > 0$, instead of e^{-x}. Show that the mean number of iterations of the rejection method is minimized when $\lambda = 1$.

Solution.

Step 1. With $f(x) = \sqrt{2/\pi}\, e^{-x^2/2}$,

$$\frac{f(x)}{g_\lambda(x)} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda x - x^2/2} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}}\, e^{\lambda^2/2} e^{-(x-\lambda)^2/2},$$

whose maximum over $x > 0$ is attained at $x = \lambda$. So the smallest admissible constant is $c(\lambda) = \sqrt{2/\pi}\, e^{\lambda^2/2}/\lambda$.

Step 2. By Theorem 10.2 the mean number of iterations is $c(\lambda)$. Now $\frac{d}{d\lambda} \log c(\lambda) = \lambda - \frac{1}{\lambda}$ is negative on $(0, 1)$ and positive on $(1, \infty)$, so c is minimized at $\lambda = 1$, with $c(1) = \sqrt{2e/\pi} \approx 1.32$. ■

Exercise 10.13 (Polar coordinates of a uniform point in the disc)

Let (X, Y) be uniform on the unit disc, with density $1/\pi$ there, and let (R, Θ) be its polar coordinates, $\Theta \in (0, 2\pi)$.

(a) Show that R and Θ are independent, $R^2 \sim U(0,1)$ and $\Theta \sim U(0, 2\pi)$.
(b) Show that if V_1, V_2 are independent $U(-1,1)$, then given $V_1^2 + V_2^2 \le 1$ the point (V_1, V_2) is uniform on the unit disc. (Together, (a) and (b) justify the polar method, Example 10.4.)

Solution.

(a)

Step 1. The map $(r,\theta) \mapsto (r\cos\theta, r\sin\theta)$ is one-to-one from $(0,1)\times(0,2\pi)$ onto the disc minus a segment of probability 0, with Jacobian r. By Theorem 6.3 applied to its inverse, (R,Θ) has density

$$f_{R,\Theta}(r,\theta) = \frac{1}{\pi}\cdot r = 2r\cdot\frac{1}{2\pi}, \qquad 0<r<1,\ 0<\theta<2\pi.$$

Step 2. The density factors, so R and Θ are independent (Proposition 6.1), with densities $2r$ on $(0,1)$ and $1/(2\pi)$ on $(0,2\pi)$. Finally $P(R^2 \le s) = P(R \le \sqrt{s}) = \int_0^{\sqrt{s}} 2r\,dr = s$ for $0<s<1$.

(b) (V_1, V_2) is uniform on the square $[-1,1]^2$ of area 4, and the disc D has area π. For $A \subseteq D$,

$$P((V_1,V_2)\in A \mid (V_1,V_2)\in D) = \frac{\text{area}(A)/4}{\pi/4} = \frac{\text{area}(A)}{\pi},$$

which is the uniform distribution on D. ■

Exercise 10.A (Antithetic variables are negatively correlated)

Let h be nondecreasing on $(0,1)$ with $E[h(U)^2] < \infty$ for $U \sim U(0,1)$. Show that $\mathrm{Cov}(h(U), h(1-U)) \le 0$.

Solution. [Supplement]

Step 1. Put $k(u) = h(1-u)$, which is nonincreasing. For all $u, v \in (0,1)$ the numbers $h(u) - h(v)$ and $k(u) - k(v)$ have opposite signs (or one is 0), so

$$\big(h(u)-h(v)\big)\big(k(u)-k(v)\big) \le 0.$$

Step 2. Let $V \sim U(0,1)$ be independent of U and substitute $u = U$, $v = V$. Taking expectations, expanding, and using that U and V have the same law and are independent,

$$0 \ge E[h(U)k(U)] - E[h(U)]E[k(V)] - E[h(V)]E[k(U)] + E[h(V)k(V)] = 2\,\mathrm{Cov}\big(h(U),k(U)\big).$$

All expectations are finite by the Cauchy–Schwarz inequality, since $k(U) = h(1-U)$ has the same law as $h(U)$.

Step 3. Hence $\mathrm{Cov}(h(U), h(1-U)) \le 0$, and the antithetic estimator $\frac{1}{2}\{h(U) + h(1-U)\}$ has variance at most $\frac{1}{2}\mathrm{Var}(h(U))$, the variance of the average of two independent copies. ■